

Project Stochastic Models in Practice

Course code: 202200339

Part of AM Module 8

2026

Redesigning a CT scan department

A radiology department in a hospital wants to completely redesign its CT scan (sub)department and asks for your advice.

First, the management wants to know how many CT scanners they should have. Along with this, they ask you to propose a fixed, weekly recurring ‘blueprint’. A blueprint is a schedule that describes per CT scanner per (weekday) day-part (i.e., morning and afternoon) whether that CT scanner is open (i.e., able to scan patients) or closed (i.e., not needing personnel, thus avoiding unnecessary costs). The hours between 8:00 and 16:00 on weekdays are called office hours, the rest of the time (i.e., 16:00-8:00 on weekdays as well as the entire Saturday and Sunday) is outside office hours. On a weekday, each scanner could be opened:

- from 8:00 till 16:00, or
- only during the morning (8:00-12:00) or
- only during the afternoon (12:00-16:00).

Outside office hours at least one scanner needs to be available. The duration of making a CT scan, in minutes, is uniformly distributed on $[10, 19]$ for all patients.

Second, the management would be interested in the effects of a ‘dynamic blueprint’. A dynamic blueprint can differ from week to week, based on the number of patients on the waiting list just before the start of the week. Hence, management asks you to develop a tool they can use to decide on Friday evening, based on the waiting list at that moment, on which day-parts each of the CT scanners should be open (or closed) during the next week. Additionally, the management asks you to provide them with insight into how performance would be if they would start using your tool, as compared to using a fixed, weekly recurring blueprint.

Third, management asks for your advice on patient scheduling. On the CT scanners, they scan three types of patients. After they have decided (on Friday evening) on a blueprint for the next week, management would like to receive your advice on when, during each day(-part), they should schedule outpatient appointments and when they should leave slots open for emergency patients and inpatients.

Emergency patients Emergency patients arrive unexpectedly and need to be scanned as soon as possible after their arrival, they thus have priority over the other two patient types. Emergency patients arrive according to a Poisson process with fixed rate $\lambda_E = 24$ per day.

Inpatients Inpatients’ requests also arrive unexpectedly. These patients are hospitalized and can thus wait for some time, but scanning them on the day of their request within office hours (i.e., between 8:00 and 16:00) is strongly preferred, as postponing their scan to a later day will likely extend their length of stay in the hospital which is highly undesirable for multiple reasons. The arrival process of the inpatient requests is an inhomogeneous Poisson process where the arrival rate is constant λ_I until 9:00 and after 15:00, and can be described by the sum of λ_I and a sine function between 9:00 and 15:00. From 9:00, the arrival rate increases from λ_I to a peak around 10:30 AM, after which the arrival rate decreases to λ_I around 12:00 noon. From 12:00 noon, the arrival rate increases again to a peak around 13:30 PM, and subsequently decreases again to λ_I around 15:00. On a weekday on average 21 inpatient requests arrive during office hours and on average 6 inpatient requests outside office hours. When an inpatient’s scan request arrives, the inpatient will stay at the nursing ward until the CT department calls the nursing ward to let them know that the inpatient can be scanned soon. The CT department makes these calls with the aim of always having (at most) one inpatient in the waiting room when there are pending inpatient requests. More specifically, if an inpatient’s request arrives while there is neither an inpatient in the waiting room nor an inpatient

being transported from a ward to the CT department, the ward is asked to send the inpatient to the CT department immediately. Otherwise, as soon as the inpatient in the waiting room starts to be scanned, the CT department calls for the next inpatient to be sent to the waiting room. After the call, it takes some time before the inpatient arrives at the CT department. This time, in minutes, is uniformly distributed on [9, 15].

Outpatients Outpatients call for an appointment, which can be scheduled at the earliest on the day after the call. On average $\lambda_O = 23$ calls from outpatients arrive on a weekday during office hours according to a Poisson process. Outpatient appointments can be scheduled at 15-minute intervals only, i.e., at 8:00, 8:15, 8:30, ..., 15:45. If an outpatient calls for an appointment and there is an appointment slot available in the same week (at the earliest on the day after the patient's call), the patient's appointment is scheduled in that slot. If there are no appointment slots available in the same week, the outpatient is put on a waiting list. On Friday evening, all outpatients on the waiting list are scheduled in appointment slots in the next week. Outpatients arrive according to their scheduled appointments, but have a probability of not showing up (without prior notification to the CT department). Each scheduled outpatient shows up with probability $p_s = 0.84$.

As specified before, management would like to receive your advice on when, during each day(-part), they should schedule outpatient appointments and when they should leave slots open for emergency patients and inpatients. More specifically, if your advice is to open CT scanner 1 next Monday the entire day, there are 32 (15-minute) slots available. Do you allow to schedule outpatient appointments in all 32 slots? Or, for example, do you block the slots between 10:00-11:00 and 14:00-15:00 such that these slots are kept free for inpatients, emergency patients, or outpatients who got delayed earlier in the day?

Fourth, management asks you to propose a dynamic policy that prescribes, given the time of day, the number of inpatient requests, and the number of outpatients and emergency patients present in the waiting room, which patient to call from the waiting room upon completion of a scan (i.e., which patient to scan next). Note that if there is an emergency patient waiting, that patient will always be the first to be called from the waiting room.

Fifth, management would like to receive your advice on the number of chairs needed in the waiting room. Upon their arrival to the CT department, patients of all three patient types wait in the waiting room. That waiting room is also used by patients who come for pre-anesthesia screening (PAS). PAS patients arrive according to a Poisson process with rate $\lambda_P = 8.3$ per hour and undergo a sequence of activities. First, a PAS patient needs to report at the secretary desk, for administrative work and to hand in a completed questionnaire. Typically, the patient then goes to the anesthetist for consultation (75% of the patients). However, based on the patient's history and the answers to the questionnaire, it could be that it is immediately clear that the patient needs some cardiopulmonary measurements (CPMs), such as an ECG or a lung test. In that case, the patient is first seen by a nurse who performs the CPM, followed by a consultation at the anesthetist (25% of the patients). 75% of these patients can leave the hospital after their consultation with the anesthetist, but in 25% of the cases, the CPM did not provide enough information and the anesthetist decides to request additional tests, such as blood tests. After the additional test, the patient does not have to go back to the anesthetist, but can leave the hospital directly. Of the PAS patients who start their 'care pathway' with consultation, 72% can leave directly afterwards, 7% only require CPM afterwards, 17% only require the additional tests afterwards, and 4% require both CPM and the additional tests (also in this sequence). Hence, six different care pathways can be distinguished for PAS patients. For the different steps that can be part of these pathways, Table 1 specifies the amount of staff that is available as well as the time it takes to serve a patient. Whenever a PAS patient needs to wait for the next step in his or her care pathway, the patient waits in the (combined CT and PAS) waiting room. Patients who do not fit within the waiting room (i.e., who cannot find an empty chair) will still wait to be scanned; this situation is very undesirable as it is not patient-friendly.

Staff type	# Staff members	Service time distribution (in minutes)
Secretary	1	Gamma(1.01, 2.6931)
Anesthetist	3	Gamma(1.03, 18.7736)
CPM nurse	2	Gamma(1.02, 24.7861)
Additional test nurse	1	Normal(10.27, 5.617) (<i>minimum 0</i>)

Table 1: Staff member numbers and service time distributions for PAS patients, where the Gamma distribution is characterized by its shape and scale parameters, and the Normal distribution by its mean and variance.

Performance indicators

The management of the CT scan department is interested in the following performance indicators:

- CT scanner utilization, with a distinction between office hours and outside office hours.
- Access times for outpatients (the number of days between the request and the actual appointment).
- Waiting times for emergency patients and for outpatients (the time between the patient's arrival in the waiting room and the start of the scan).
- The fraction of (CT) patients who have to wait outside the waiting room.
- For inpatients with requests during office hours, the percentage of those that cannot be scanned before 16:00 on the same weekday.

Requirements

Your assignment is to formulate analytical mathematical models (i.e., no simulation models) for each of the five questions posed by the management, except the third question¹. Subsequently, use computer simulation to provide management with insight on how your advice for the different questions plays out in practice when combined in the redesign of the CT scan department.

More specifically, first formulate an analytical mathematical model per question. That is, motivate your modeling approach, describe your mathematical model including assumptions, and give its numerical results. Note that computer code is not a mathematical model. How you describe your mathematical model depends on the type of model. For example, a Stochastic Dynamic Programming model is described by its decision epochs, states, actions, immediate rewards, transition probabilities, and (recurrence relation of the) optimal value function; a queueing model is described using Kendall's notation along with the relevant parameter values. To obtain numerical results for your analytical mathematical models, you will in some cases have to write computer code. You are strongly advised to fully write down your mathematical model before you start coding.

Of course, the result (or policy) of the first question influences the result of the second question and so on. You are requested to provide a complete final advice that covers all five questions. The quality of your advice is assessed based on all performance indicators listed above. Here, you can re-use the simulation model that you will develop in the course Stochastic Simulation (as final assignment for that course). Your simulation model – with adjustments – serves two purposes:

1. To be able to formulate analytical mathematical models for each of the five questions, you have to make strong assumptions in certain cases. Simulation enables you to test the results obtained from your mathematical model in a more realistic setting.
2. By including the results (or policy) from your five analytical models into your simulation model, you can demonstrate what the combined effect of your advice would be for the practical setting of the CT scan department.

The deliverables on which you will be assessed are a *technical* report describing your formulated models and obtained results, and a presentation addressed to the *management* of the CT scan department in which you present your advice.

To help you get started with formulating analytical models for each of the five questions posed by the management, below are two specific assignments that each help you get started with one of the questions. Formulating the mathematical model and obtaining numerical results for it will help you to answer these two questions. For the remaining questions posed by management, it is up to yourself to formulate appropriate models, and to obtain numerical results for those. Note that in order to be doable (while you are still learning about the relevant models), the assignments below do contain certain (simplifying) assumptions that may not fully align with your eventual advice. The advice is that you initially work with these assumptions for the specific assignments, but you may of course get rid of these assumptions while working on your eventual advice.

¹You must also answer the management's third question, but for this question you have a choice: you may formulate an analytical mathematical model or you may apply simulation-optimization.

Specific assignment for Question 4: who to scan next?

As specified before, any assumptions listed here (and in the specific assignment for Question 5) that come on top of what has been described before are in no way binding, you have the freedom to change those whenever you like.

Assume that there is one CT scanner that is open from 8:00 to 16:00, so it has 32 (15-minute) slots available during one day. Each slot is either filled with an outpatient appointment, or left empty to accommodate a potential emergency patient or inpatient. Also assume that a CT scan has a deterministic duration of 15 minutes (i.e., one slot). Hence, at the beginning of each slot, it has to be decided which patient from the waiting room should be scanned next. (Unless there is an emergency patient waiting, because he or she will automatically be the next patient to be scanned.) At the start of the day, the waiting room is empty. Outpatients arrive according to their scheduled appointments, but have a probability $1 - p_s = 0.16$ of not showing up. Assume that, during one slot, at most one emergency patient and at most one inpatient request can arrive, independently of one another, with respective probabilities p_E and p_I . Scanning an outpatient (inpatient) gives a revenue r_O (r_I). Letting an outpatient or an inpatient request wait incurs a waiting cost per slot of w_O and w_I , respectively. At the end of office hours (i.e., after slot 32), there is a penalty per patient who is still waiting to be scanned, which is π_O per outpatient and π_I per inpatient. (These patients will be scanned on a CT scanner that stays available outside office hours.)

Formulate an SDP that can take different appointment schedules as input (i.e., 32 slots, part of which filled with a scheduled outpatient appointment). At the beginning of each slot, the SDP should suggest which patient type to admit into service, with the aim of maximizing the total profit over the 32 slots. In terms of parameter values, p_E and p_I can be calculated based on the information provided under ‘Redesigning a CT scan department’. For the revenue and cost parameters, you have the freedom to choose your own values, and can experiment with different combinations of values. Motivate your choices. You may start with $r_O = 100$, $r_I = 20$, $w_O = 1.5$, $w_I = 0$, $\pi_O = 10$, $\pi_I = 200$.

Specific assignment for Question 5: how many chairs in the waiting room?

Assume that all treatment times (for CT and PAS patients) follow exponential distributions. Develop a queueing model to determine the number of chairs needed in the waiting room, such that the probability that a patient cannot find an empty chair is low.

Suggested timeline

Dates	Activities
Weeks 1 - 6	1. Complete the specific assignment for Question 4
<i>Apr 23 - May 29*</i>	2. Develop a tool to support the management in working with a dynamic blueprint
Weeks 7 - 10.5	3. Develop an advice for how many CT scanners the department should have
<i>May 30 - Jun 24**</i>	4. Complete the specific assignment for Question 5
Weeks 10.5 - 11	5. Develop an advice on patient scheduling
<i>Jun 25 - Jul 2</i>	6. Integrate all the above in your simulation model
	7. Write your technical report
	8. Prepare your management presentation

* If you hand in your work via Canvas on or before May 29, you can get intermediate feedback.

** If you hand in your work via Canvas on or before June 24, you can get intermediate feedback.